RESTRICTION OF SCALARS ALONG A FINITE LOCALLY FREE MORPHISM IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Declared dependency: lemma-fibre-diagram carries an 'Omitted. Hint: Exercise' proof in the source; used here as a declared reference only (external_references), not re-proved.

Allowed standard prerequisites (editorial).

- Mor functor algebraicity. proposition-hom-functor-algebraic-space (same chapter, named). Algebraicity of Mor_B(Z, X) and Mor_B(Z, Z).
- Cartesian identification. lemma-fibre-diagram (same chapter, named; source proof 'Omitted. Hint: Exercise'). Identifies Res_{Z/B}(X) as the fibre product Mor_B(Z,X) x_{Mor_B(Z,Z)} B.
- Fibre products of spaces. spaces-lemma-fibre-product-spaces-over-sheaf-with-representable-diagonal (cross-chapter, named). Existence of the fibre product.

Source reference. Original lemma 05YE: sources/stacks-criteria/pinned-original-source.tex, source lines 1891–1891. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 05Y7: sources/stacks-criteria/pinned-original-source.tex, source lines 1642–1642. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04T9: focused-seeds/geometry/S9-crit-restriction-of-scalars-algebraic-space/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let $X \to Z \to B$ be morphisms of algebraic spaces over S. If $Z \to B$ is finite locally free then $Res_{Z/B}(X)$ is an algebraic space.*

Proof. By Proposition 05Y7 the functors $Mor_B(Z, X)$ and $Mor_B(Z, Z)$ are algebraic spaces. Hence this follows from the cartesian diagram of Lemma 05YE and the fact that fibre products of algebraic spaces exist and are given by the fibre product in the underlying category of sheaves of sets (see Spaces, Lemma 04T9). □

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